

Untargeted Metagenomics - Taxa and Functional Annotation

Meta_Microbial Workshop

Metagenomic and bioinformatic insights into
microbial communities

 @AdrianaRego10

 adrianairego@gmail.com



Sponsorship

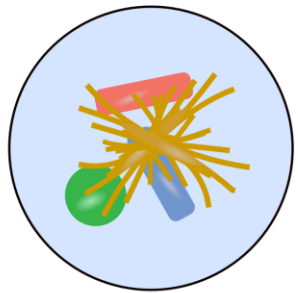


1

Introduction

Amplicon vs shotgun metagenomics

Mixed microbial community



DNA
Extraction



Amplicon sequencing



Multiple copies of fragments
from 1 target gene

Metagenomics sequencing



Short sequence
fragments from "all" DNA

PCR amplification of marker genes
Quick and cheap analysis
Applicable to samples contaminated with host DNA

Limitations

Only functional predictions (e.g. PICRUST)
Limited taxonomic resolution
PCR and primer bias

Sequencing of short/long random DNA fragments
Functional potential
Uncultured genomes
Taxonomic resolution (species/strain level)

Limitations

Expensive, time-consuming and computationally
intense analysis
Host contamination

<https://astrobiomike.github.io/>

Amplicon vs shotgun metagenomics

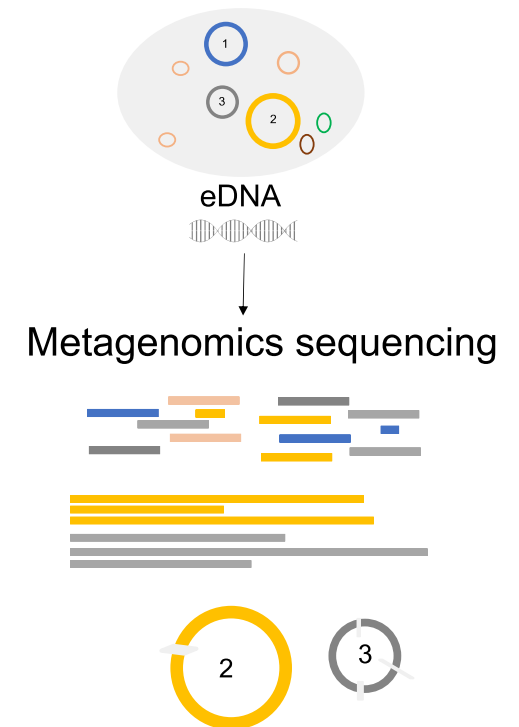
Shotgun metagenomics

Strengths

- High taxonomical resolution (species/strains)
- Functional metagenomics (genes)
- Comparative genomics

Challenges and limitations

- Completeness of databases and good references
- DNA extraction bias (cell-lysis efficiency)
- No information on expression of genes (transcriptomics instead)
- Complexity of ecosystems and sequencing depth
- Contamination



1

Introduction

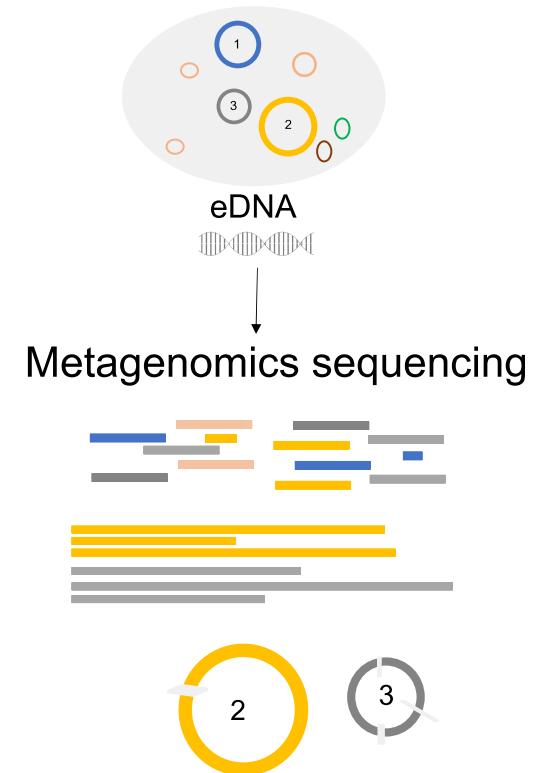
Shotgun sequencing metagenomics

Which questions can we answer using metagenomics data?

Who is there? - Taxonomical annotation

What can they do? - Functional annotation

And not “What are they doing?” -> Transcriptomics

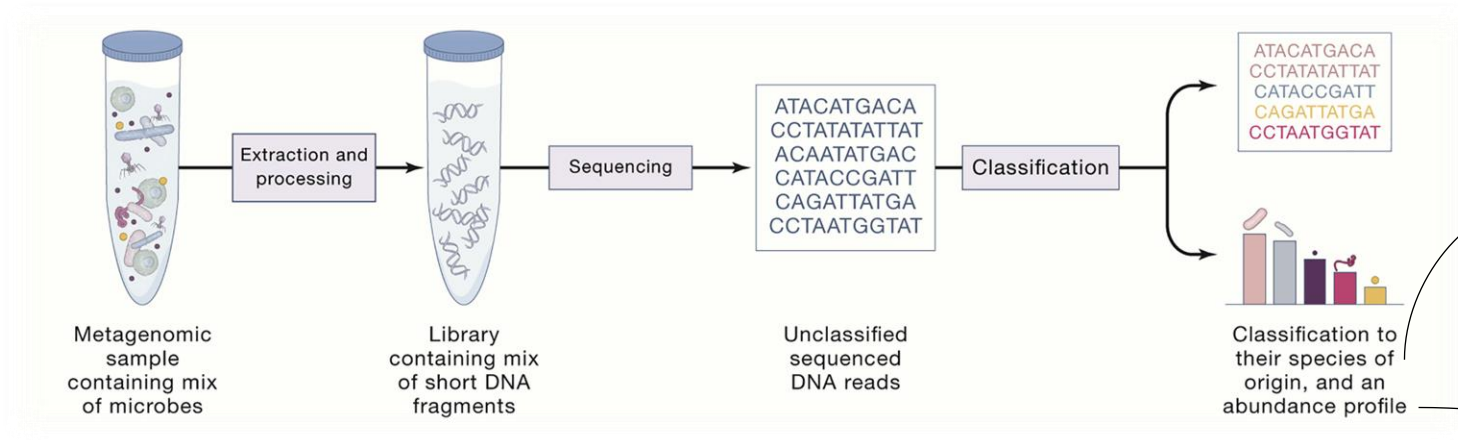


2

Taxonomical annotation

Who is there?

MAGs Taxonomic Classification and Profiling



Taxonomic classification/binning

Assigns taxon labels to individual sequences and results in taxon-specific sequence bins (and sequence abundance profiles).

Taxonomic profiling

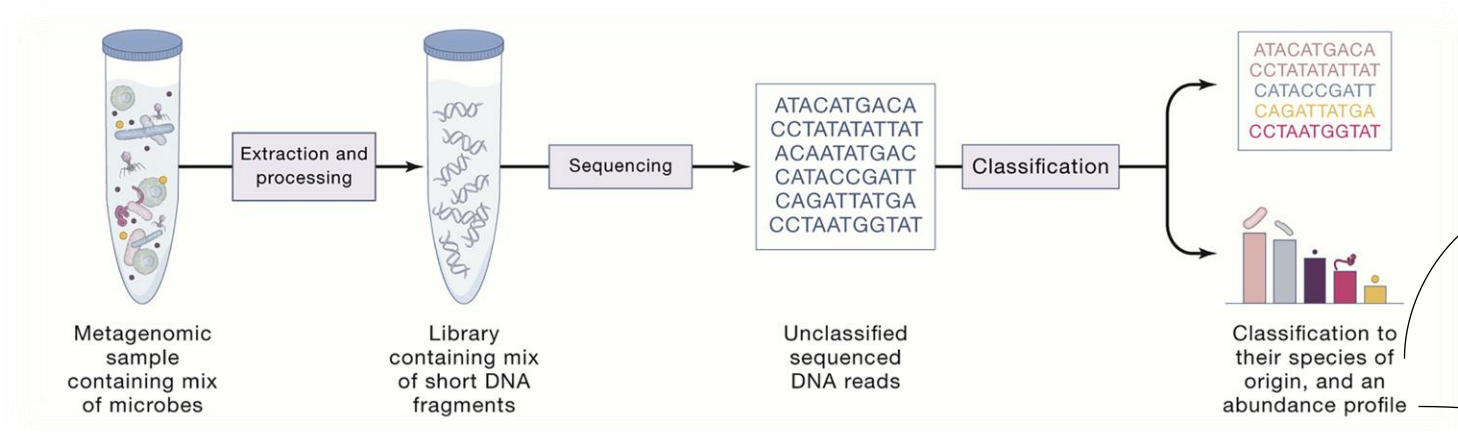
Quantify the presence and relative taxon abundances of microbial communities from metagenome samples.

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Quantify the presence and relative taxon abundances of microbial communities from metagenome samples.

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Taxonomical annotation

Who is there?

Taxonomic classifiers

Traditional methods based on **16S rRNA** applied to metagenomics - limited resolution and 16S rRNAs can be poorly represented in MAGs.

Instead the use of several **single-copy marker genes** improves resolution.

Identification of marker genes in
MAGs/metagenomes

MSA with marker genes from
reference databases

Phylogenetic tree
ANI – average nucleotide identity

2

Taxonomical annotation

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Calculation of average nucleotide identity (ANI) between genomes is used for **species identification**. ANI >96% - species
ANI of <93% - species differentiation.

2

Taxonomical annotation

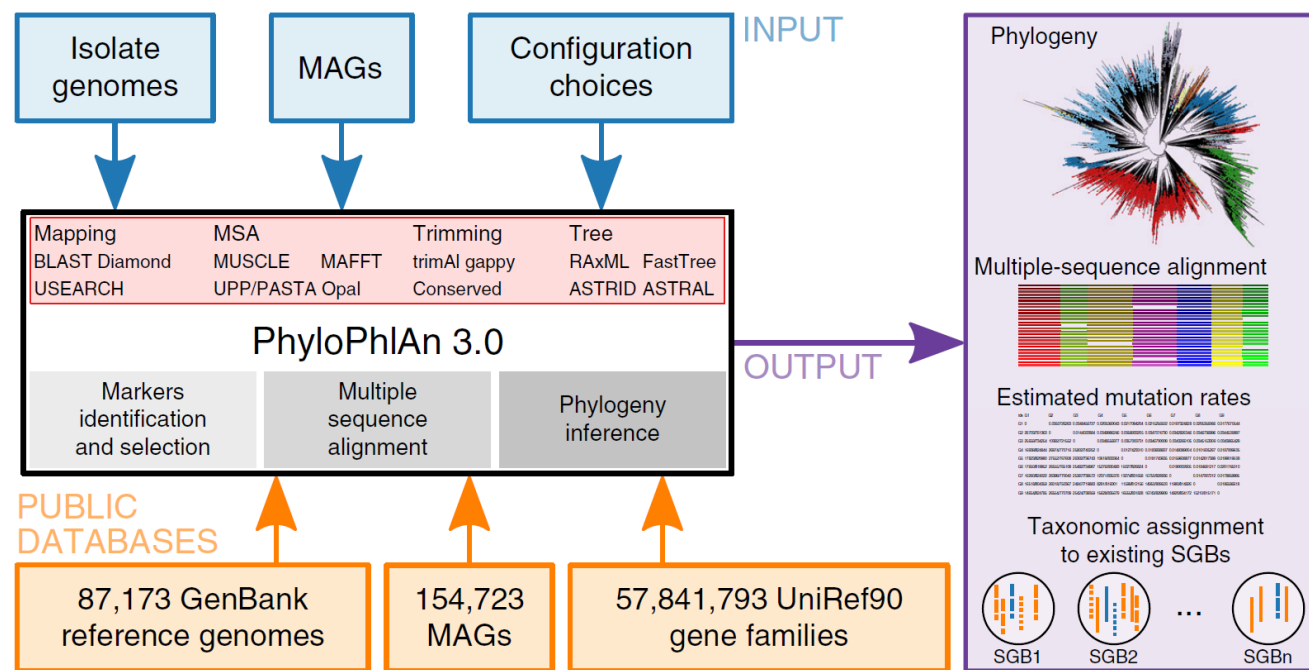
Who is there?

Taxonomic classifiers

PhyloPhlAn -

<https://huttenhower.sph.harvard.edu/phylophlan/>

is an integrated pipeline for large-scale phylogenetic profiling of genomes and metagenomes. PhyloPhlAn is an accurate, rapid, and easy-to-use method for large-scale microbial genome characterization and phylogenetic analysis at multiple levels of resolution. PhyloPhlAn can assign both genomes and metagenome-assembled genomes (MAGs) to species-level genome bins (SGBs).



(Adapted from Asnicar et al. 2020)

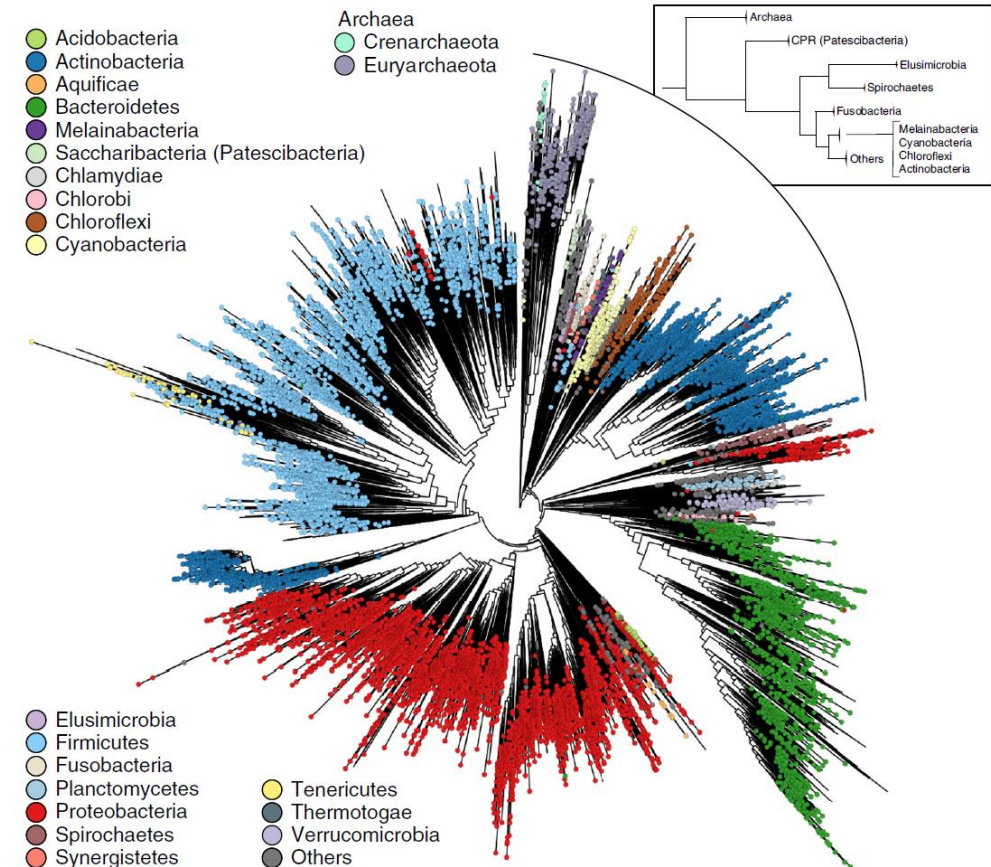
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Taxonomical annotation

Who is there?

Taxonomic classifiers

GTDB-Tk <https://github.com/Ecogenomics/GTDBTk>

- is a software toolkit for assigning objective taxonomic classifications to bacterial and archaeal genomes based on the Genome Database Taxonomy.
- command line version



GTDB R214 spans 402,709 genomes organized into 85,205 species clusters.

	Bacteria	Archaea	Total
Phylum	161	20	181
Class	488	60	548
Order	1,624	148	1,772
Family	4,262	508	4,772
Genus	19,153	1,586	20,739
Species	80,789	4,416	85,205

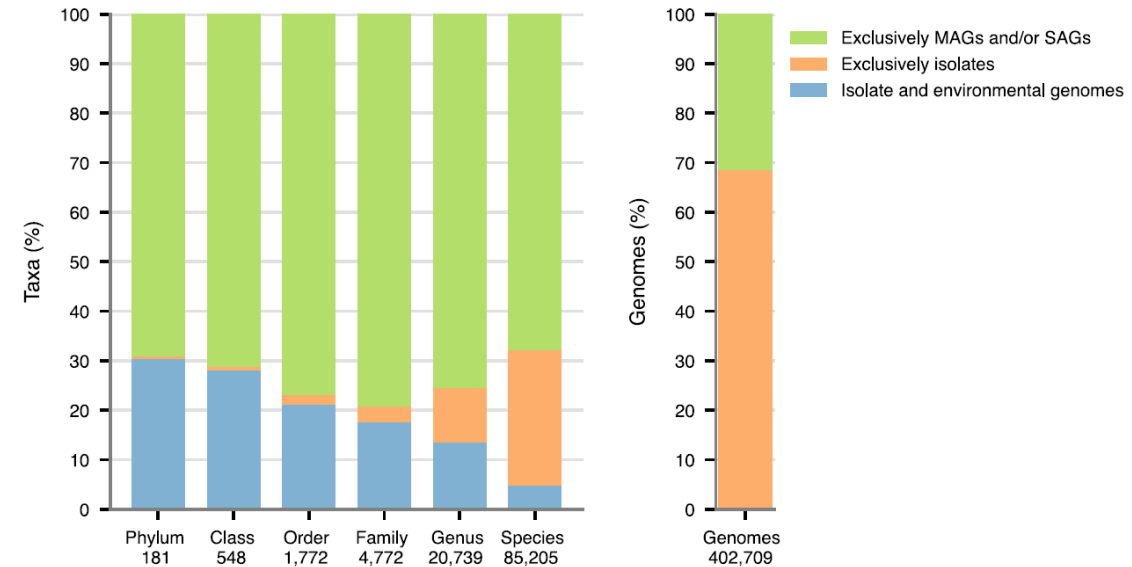
```
gtdbtk classify_wf --genome_dir All_bins/ --out_dir  
All_bins_gtdbk_output -x fa --cpus 50
```

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```

2

Taxonomical annotation

Who is there?

Taxonomic classifiers

Microbial Genome Atlas

<http://microbial-genomes.org/>

TypeMat

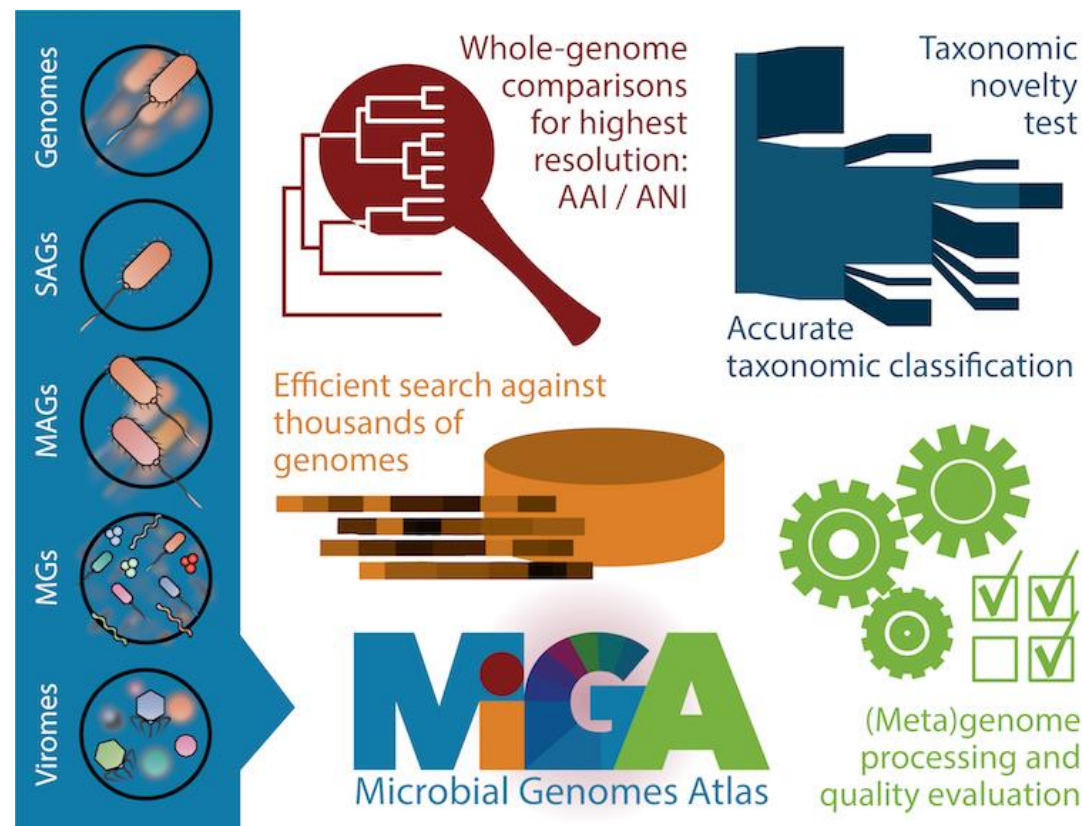
Query the collection of type material genomes to identify the closest genomes available from formally named species and determine taxonomic classification and novelty rank

Upload genome Explore data

NCBI Prok

Query the complete- and chromosome-level NCBI Genome database (Prokaryotes) with your own genomes to identify the most closely related complete genomes available

Upload genome Explore data



Who is there?

Taxonomic classifiers

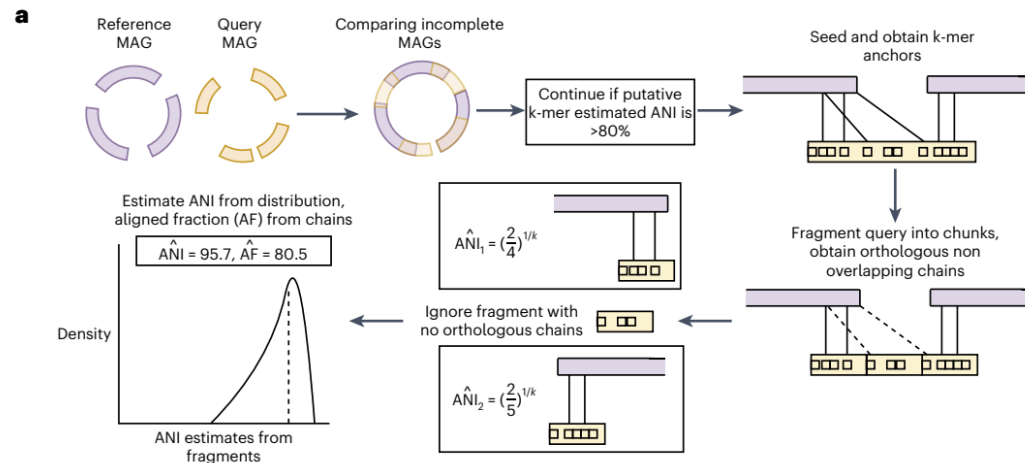
ANI tools

Comparison of ANI against a specific (even a created) database

FastANI - <https://github.com/ParBLiSS/FastANI>

Skani – <https://github.com/bluenote-1577/skani>

performs better for incomplete MAGs (e.g. for metagenor environmental studies)

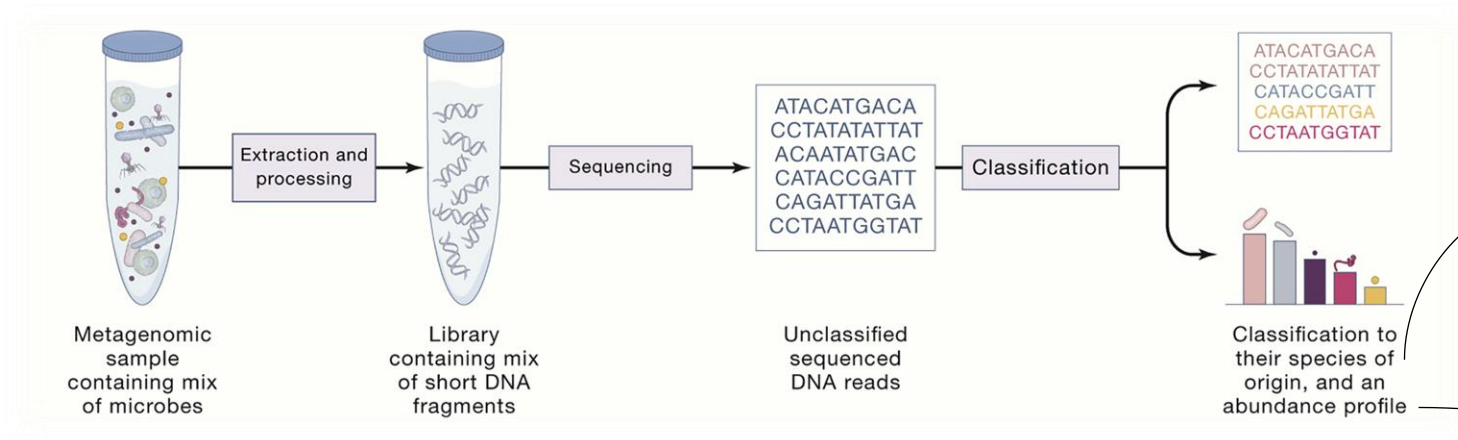


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Taxonomical annotation

Who is there?

Metagenomic profilers

Metagenomic profilers can be categorized based on their reference database type.

Reference
database

Software

Approaches for Taxonomic classification and profiling

DNA to DNA

DNA-to-DNA
Complete genomes as reference



Kraken 2

DNA to protein

DNA-to-protein
Coding DNA as reference



Diamond
Kaiju

Marker gene based

DNA-to-marker
Single copy markers as reference



MetaPhlan
mOTUs

2

Taxonomical annotation

Who is there?

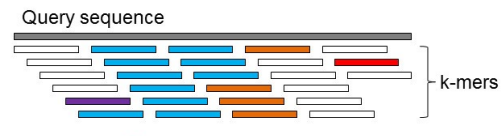
Metagenomic profilers

The **comparison** of reads to database sequences can be performed by different strategies.

Approaches for Taxonomic classification and profiling

K-mer based

Very fast
Lack sensitivity
Big fraction unclassified



Kraken 2

Marker gene based

Estimation of species
abundance
Low detection accuracy
and that the unclassified
percentage is unknown

MetaPhlan

Genome based

High detection accuracy,
unclassified percentage
is known
High-resolution genomic
comparisons are possible

GTDB-Tk

Software

2

Taxonomical annotation

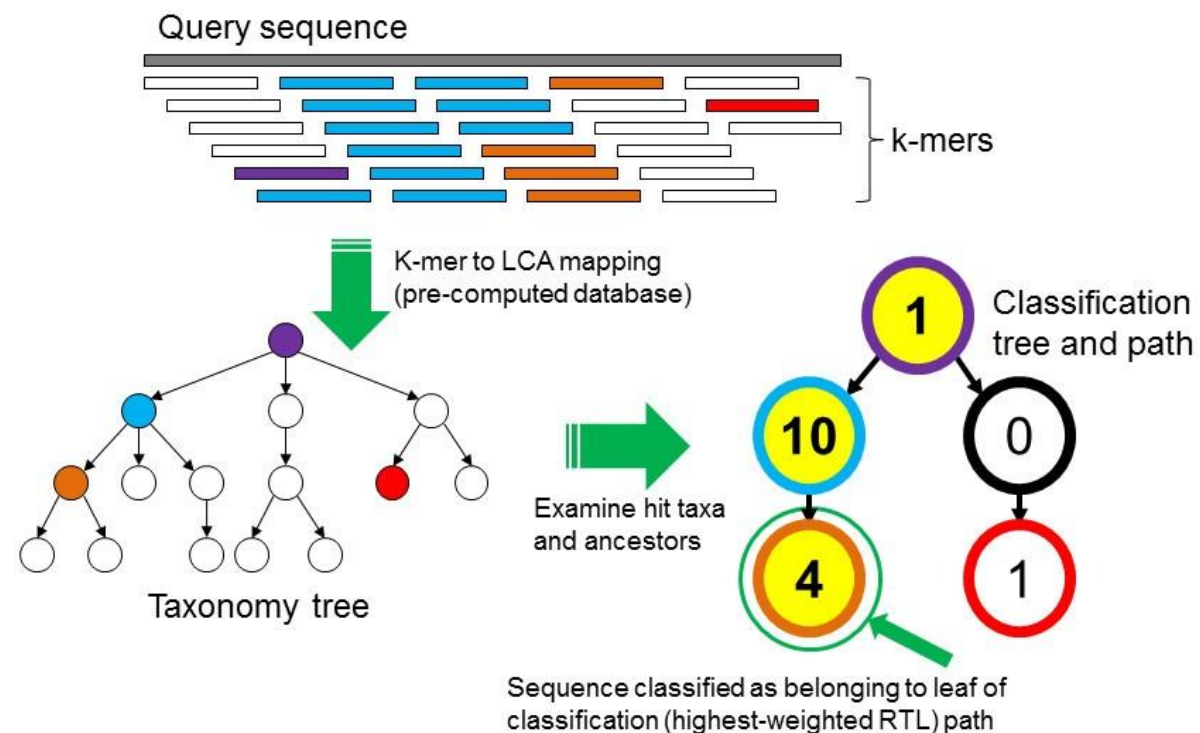
Who is there?

Kraken 2 - <https://ccb.jhu.edu/software/kraken2/>

Kraken is a system for assigning taxonomic labels to short DNA sequences, usually obtained through metagenomic studies.

Reference database – DNA to DNA
Software – kmer-based

***MiniKraken:** To allow users with low-memory computing environments



Classification

To classify a set of sequences (reads), use the kraken command:

```
kraken --db $DBNAME seqs.fa
```

Who is there?

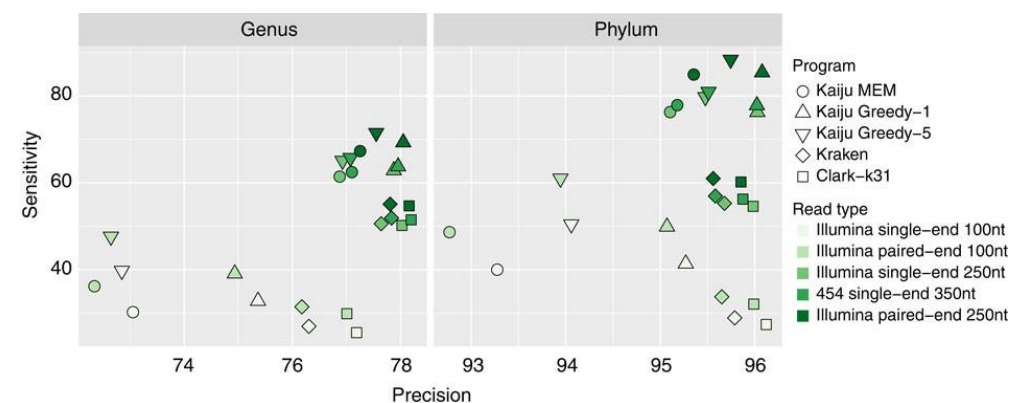
Kaiju - <https://bioinformatics-centre.github.io/kaiju/>

Kaiju is a program for fast and sensitive taxonomic classification of high-throughput sequencing reads from metagenomic, whole genome sequencing or metatranscriptomics experiments.

Reference database – DNA to protein



Fast and sensitive taxonomic classification for metagenomics



(Adapted from Menzel et al. 2016)

2

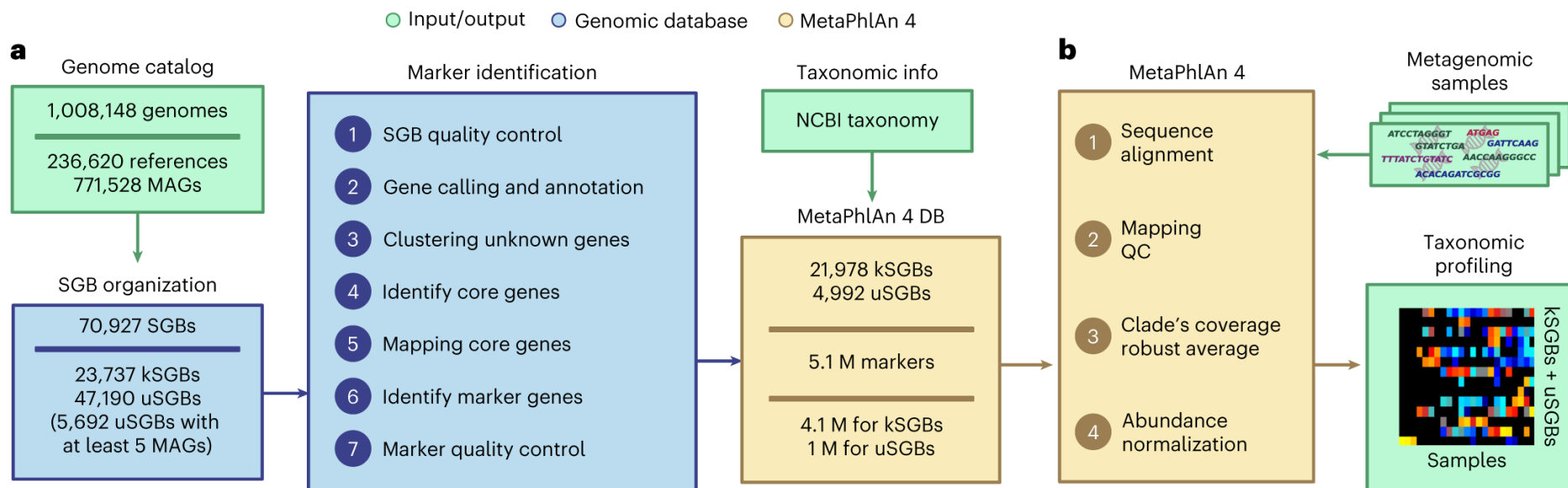
Taxonomical annotation

Who is there?

Software – marker gene based

MetaPhlAn: Metagenomic Phylogenetic Analysis

- is a computational tool for profiling the composition of microbial communities with **species-level**. With **StrainPhlAn**, it is possible to perform accurate strain-level microbial profiling.



MetaPhlAn 4 relies on ~**5.1M** unique clade-specific marker genes identified from ~**1M** microbial genomes (~236,600 references and 771,500 metagenomic assembled genomes) spanning 26,970 species-level genome bins (SGBs).

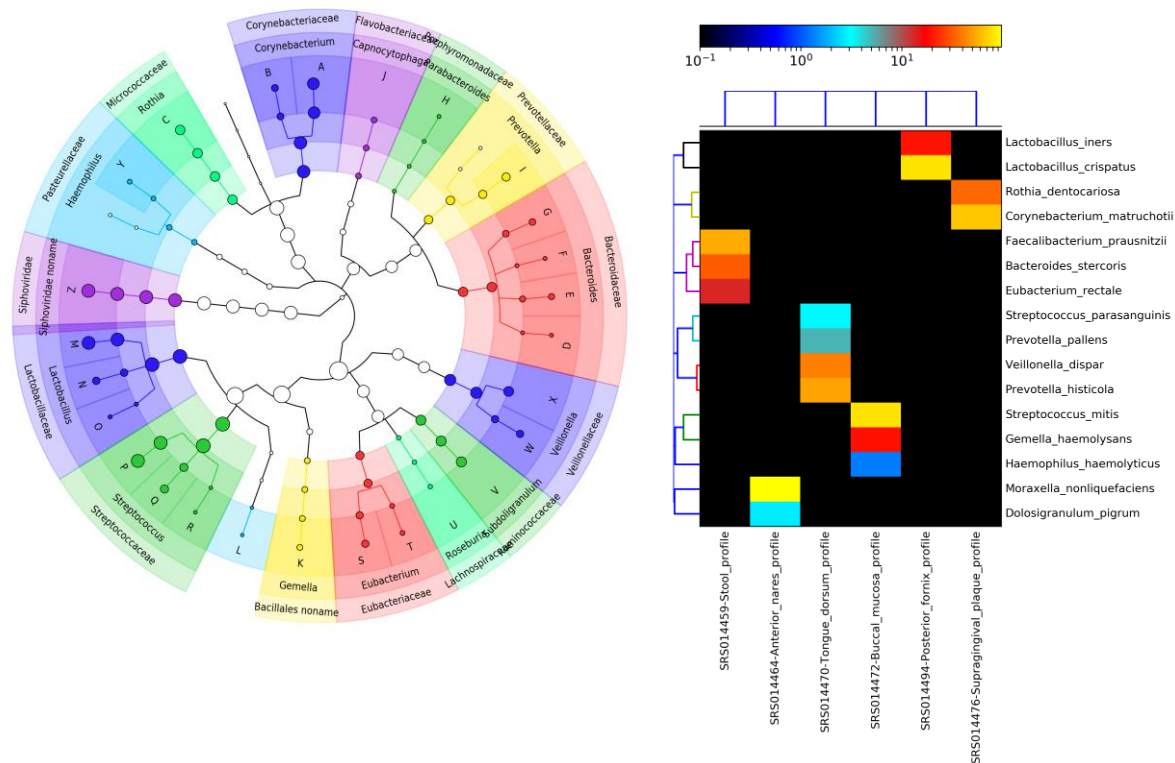
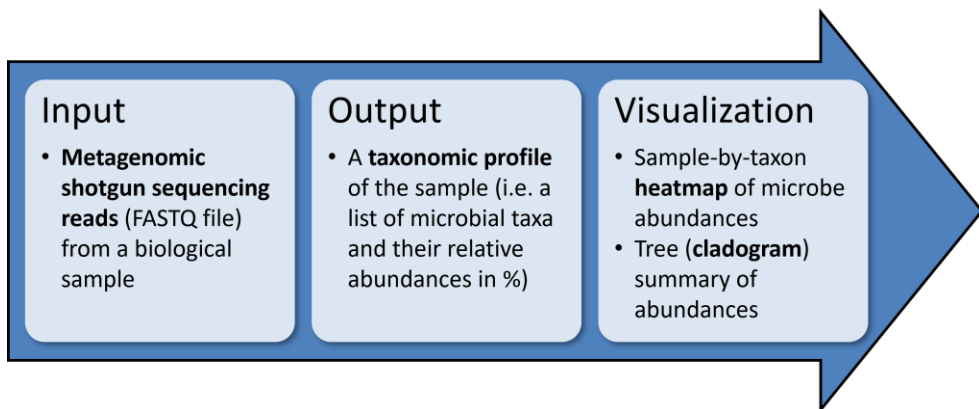
2

Taxonomical annotation

Who is there?

Software – marker gene based

MetaPhlAn: Metagenomic Phylogenetic Analysis



Examples visualization

(<https://github.com/biobakery/biobakery/wiki/metaphlan4>)

2

Taxonomical annotation

Who is there?

Benchmarking of existent tools for taxonomic classification

How to choose the best tool?

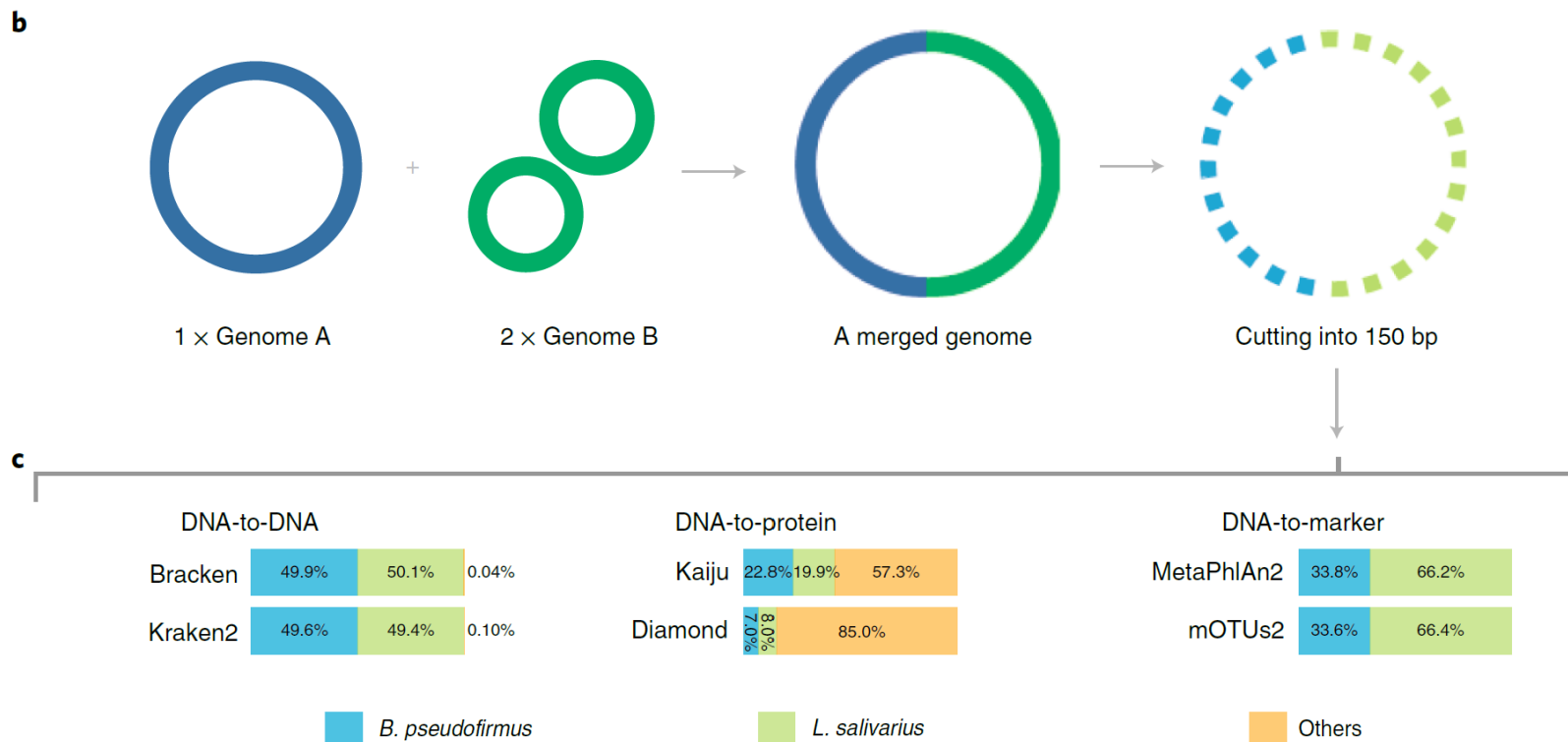
- **Question** - do I want to classify a specific MAG or to have an overview of the taxonomical diversity of a dataset? Classify a potential new genus, species?
- **Sample type** –completeness of reference databases
- **Availability** of computational resources and time

2

Taxonomical
annotation

Who is there?

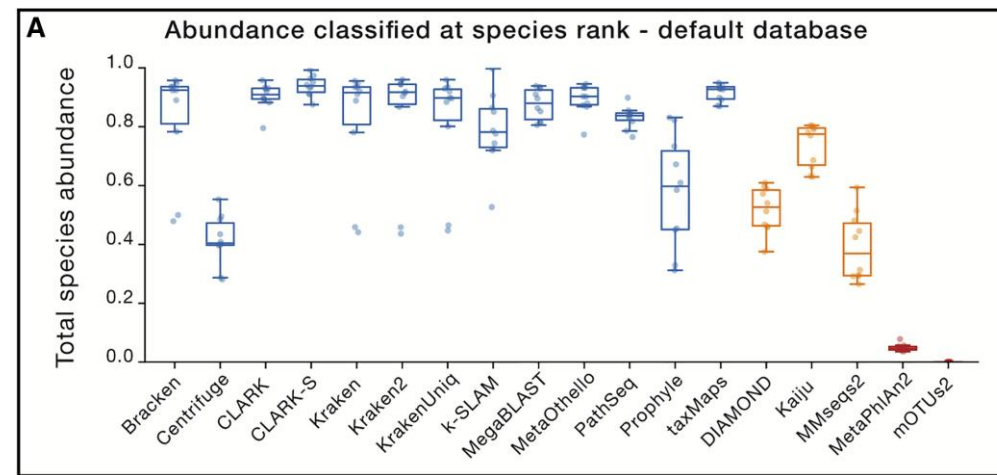
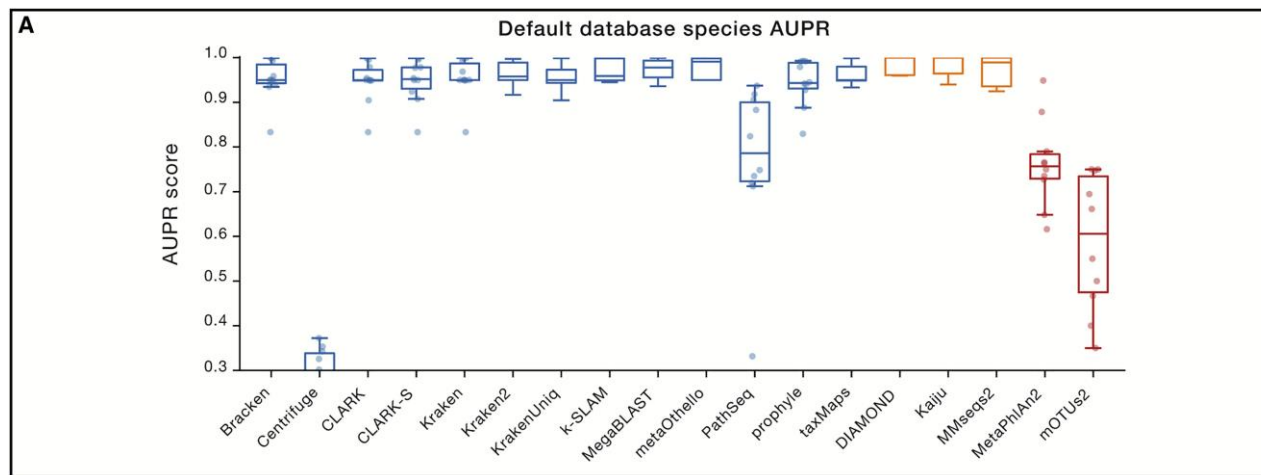
Benchmarking of existent tools for taxonomic classification



(Adapted from Sun et al. 2021)

Who is there?

Benchmarking of existent tools for taxonomic classification



— DNA-to-DNA — DNA-to-Protein — DNA markers

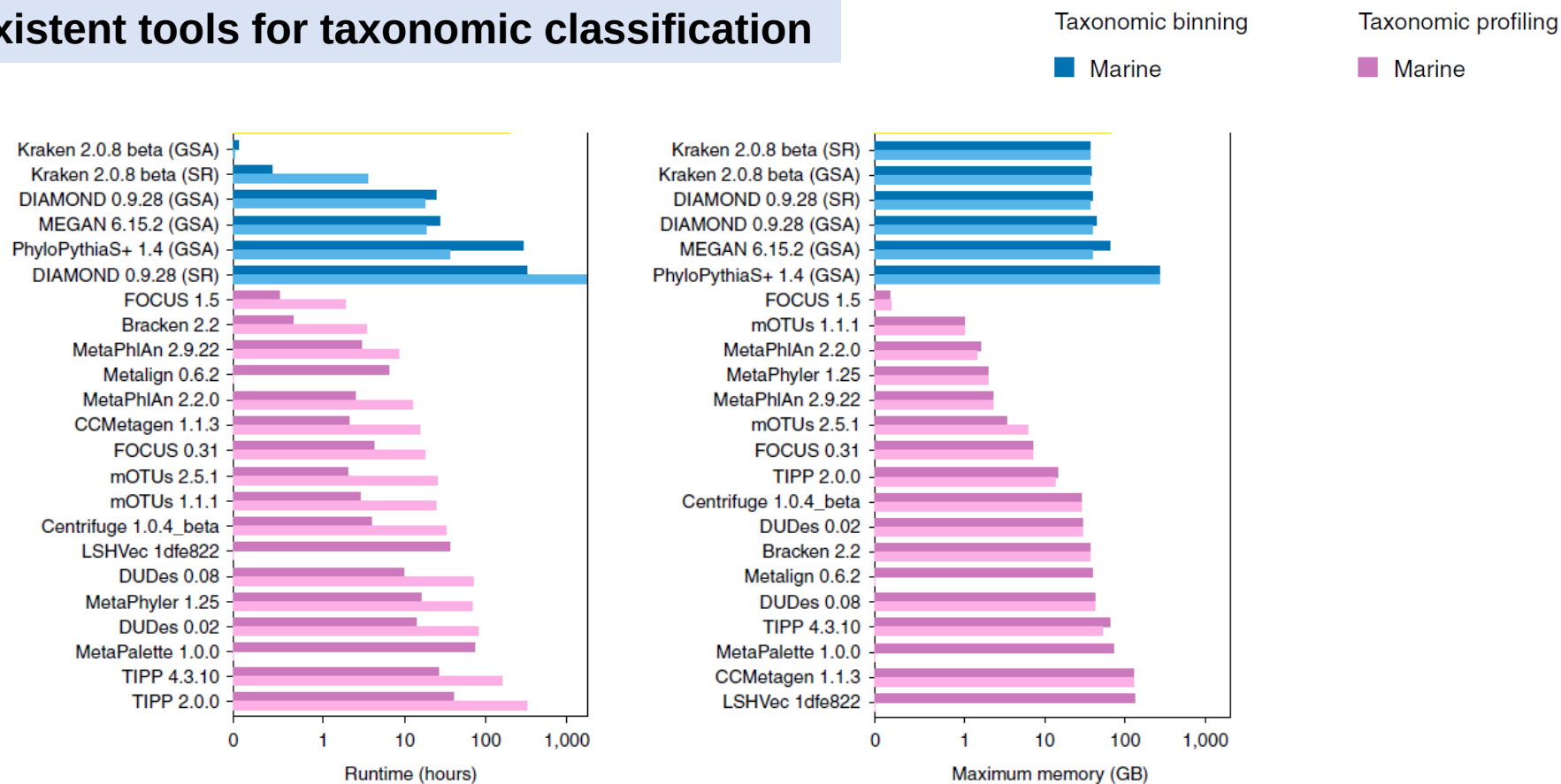
(Adapted from Ye et al. 2019)

2

Taxonomical annotation

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Benchmarking of existent tools for taxonomic classification

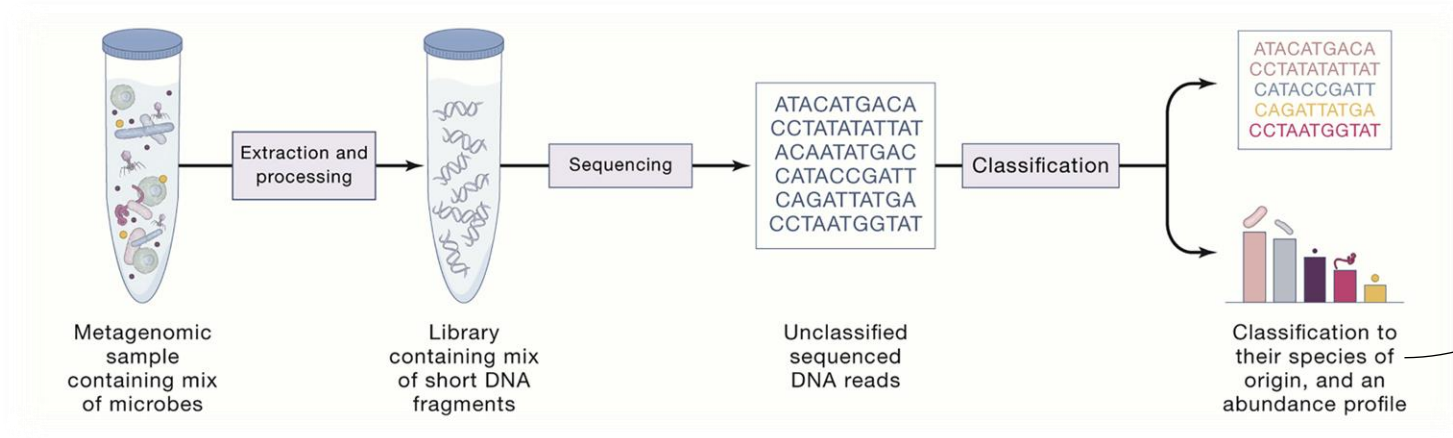


(Adapted from Meyer et al. 2022)

3

Functional annotation

What can they do?



Functional annotation

What can the identified microbial community (potentially) do?

→ **function prediction** for protein coding gene

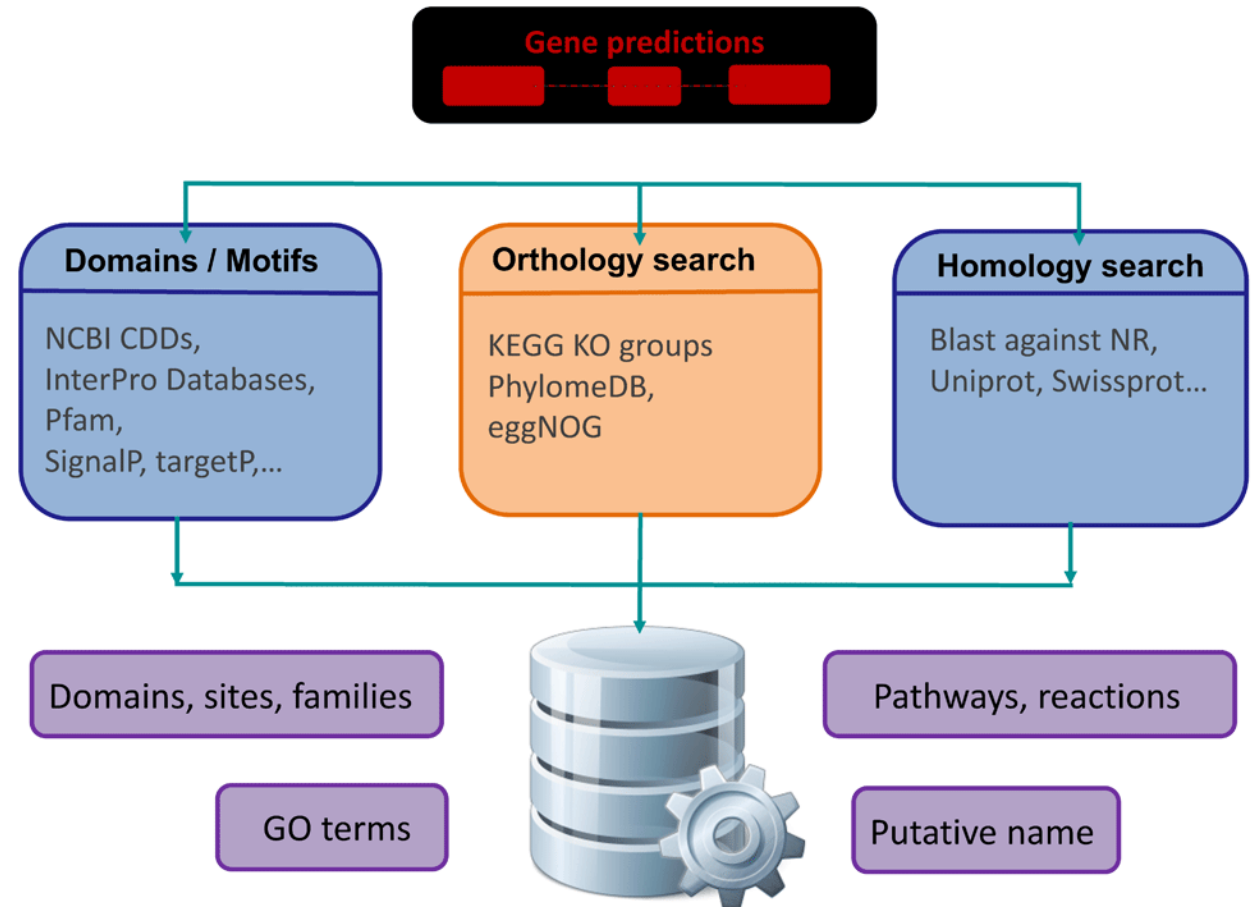
3

Functional annotation

What can they do?

Gene functional annotation tools can be classified into two categories:

- 1) tools with broad scopes to evaluate **full functional potential**
- 2) tools with narrow scopes focusing on one or a few **specific biological processes**.



(Adapted from Angel et al. 2018)

3

Functional annotation

What can they do?

Databases for Functional Annotation

KEGG: Kyoto Encyclopedia of Genes and Genomes

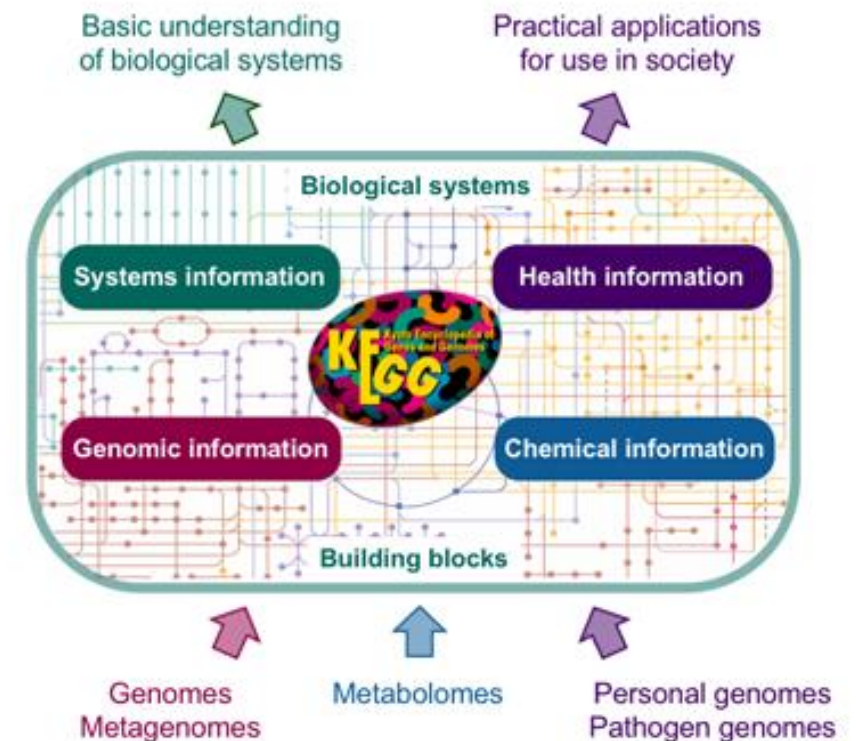
KEGG is a database resource for understanding high-level functions and utilities of the biological system.

- computer representation of the biological system, consisting of molecular building blocks of genes and proteins (**genomic information**) and chemical substances (**chemical information**) that are integrated with the knowledge on molecular wiring diagrams of interaction, reaction and relation networks (systems information). It also contains disease and drug information (**health information**) as perturbations to the biological system.



KEGG PATHWAY Database

Wiring diagrams of molecular interactions, reactions and relations



3

Functional annotation

**What can they
do?**

Databases for Functional Annotation

**COG - Database of Clusters of Orthologous Genes
(COGs)**

<https://www.ncbi.nlm.nih.gov/research/cog>

COGs	Genomic loci	Taxonomic Categories	Organisms	Protein IDs	COG symbols
4,877	3,456,089	37	1,309	3,213,255	3,828

3

Functional annotation

What can they do?

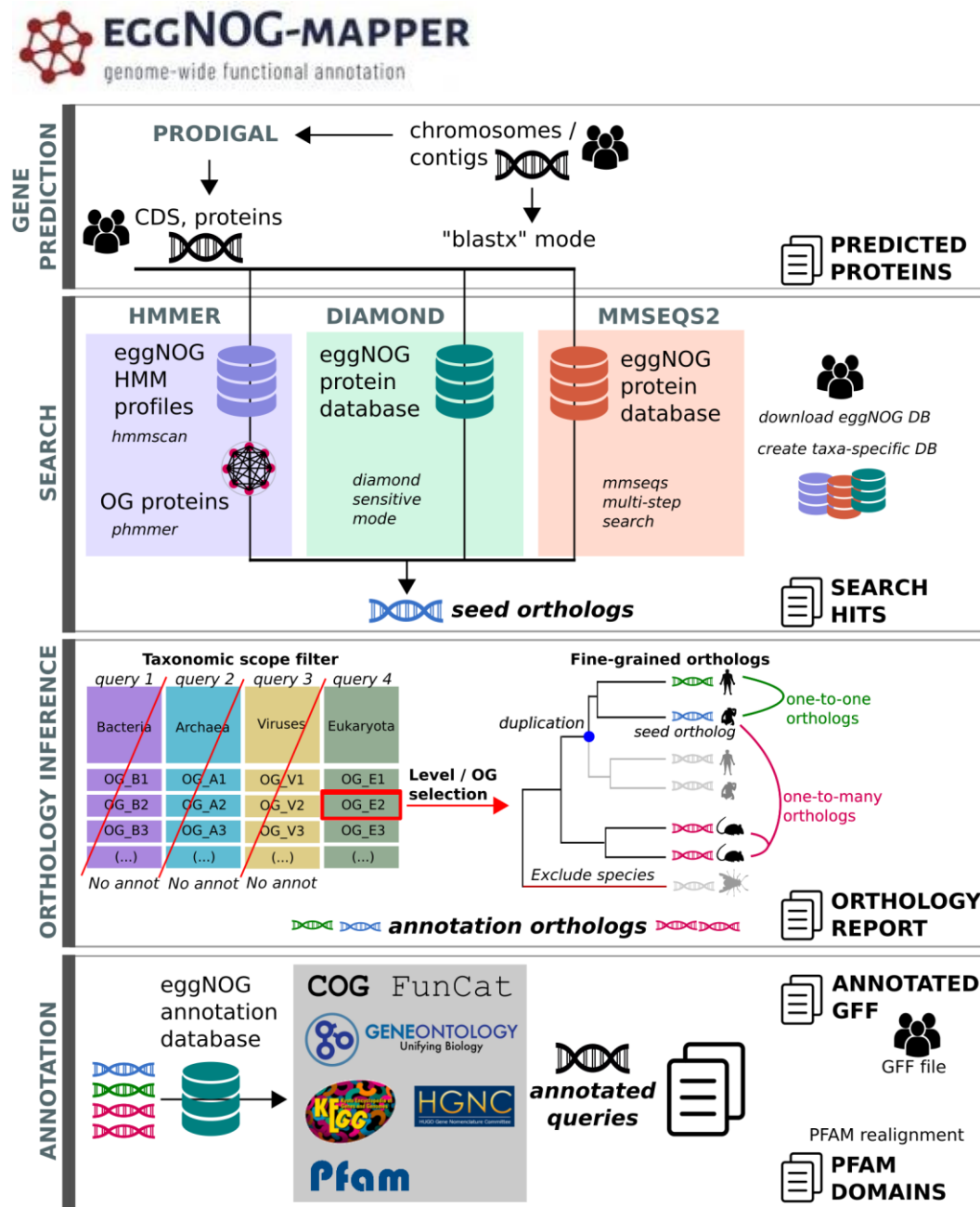
Databases for Functional Annotation

EggNOG <http://eggnog5.embl.de/#/app/home>

A database of orthology relationships, functional annotation and gene evolutionary histories, based on 5090 organisms and 2502 viruses.

Software for Functional Annotation

EggNOG-mapper - tool for fast functional annotation of novel sequences. It uses precomputed Orthologous Groups (OGs) and phylogenies from the EggNOG database.



What can they do?

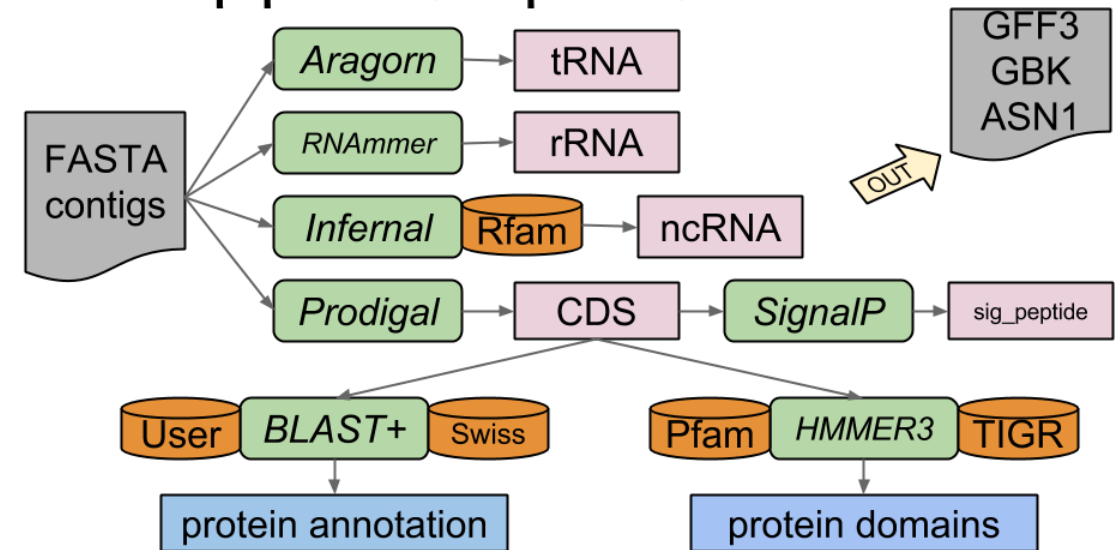
Software for Functional Annotation

Prokka: rapid prokaryotic genome annotation

<https://github.com/tseemann/prokka>

is a pipeline that runs several other tools to annotate prokaryotic genomes.

Prokka pipeline (simplified)



```
# Choose the names of the output files
% prokka --outdir mydir --prefix mygenome contigs.fa
```

3

Functional annotation

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Software for Functional Annotation

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locus_tag	ftype	length_bp	gene	EC_number	COG	product	
OCAFAAEJ_00	CDS	1620				hypothetical protein	
OCAFAAEJ_00	CDS	597				hypothetical protein	
OCAFAAEJ_00	CDS	1059				hypothetical protein	
OCAFAAEJ_00	CDS	474				hypothetical protein	
OCAFAAEJ_00	CDS	456				hypothetical protein	
OCAFAAEJ_00	CDS	786				hypothetical protein	
OCAFAAEJ_00	CDS	888				hypothetical protein	
OCAFAAEJ_00	CDS	969	dusB	1.3.1.-	COG0042	tRNA-dihydrouridine synthase B	
OCAFAAEJ_00	CDS	549	yvqK	2.5.1.17	COG2096	Corrinoid adenosyltransferase	
OCAFAAEJ_00	CDS	366				hypothetical protein	
OCAFAAEJ_00	CDS	939				hypothetical protein	
OCAFAAEJ_00	CDS	1830	btuB_1			Vitamin B12 transporter BtuB	
OCAFAAEJ_00	CDS	195	def_1	3.5.1.88		Peptide deformylase	
OCAFAAEJ_00	CDS	96				hypothetical protein	
OCAFAAEJ_00	CDS	2166				hypothetical protein	
OCAFAAEJ_00	CDS	171				hypothetical protein	

```
# Choose the names of the output files
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```

What can they do?

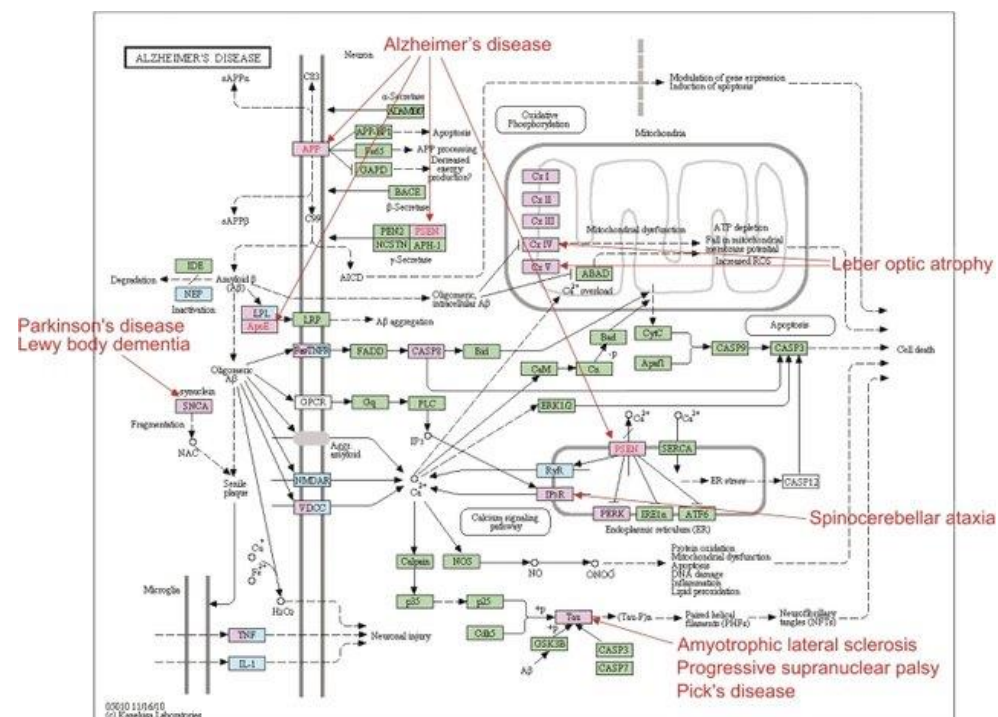
Visualization of Functional Annotation

KEGG Mapper

The Reconstruct tool "reconstructs" **KEGG pathway maps** and other network entities from a set of K numbers (KO identifiers).



KEGG Mapper – Reconstruct



3

Functional annotation

What can they do?

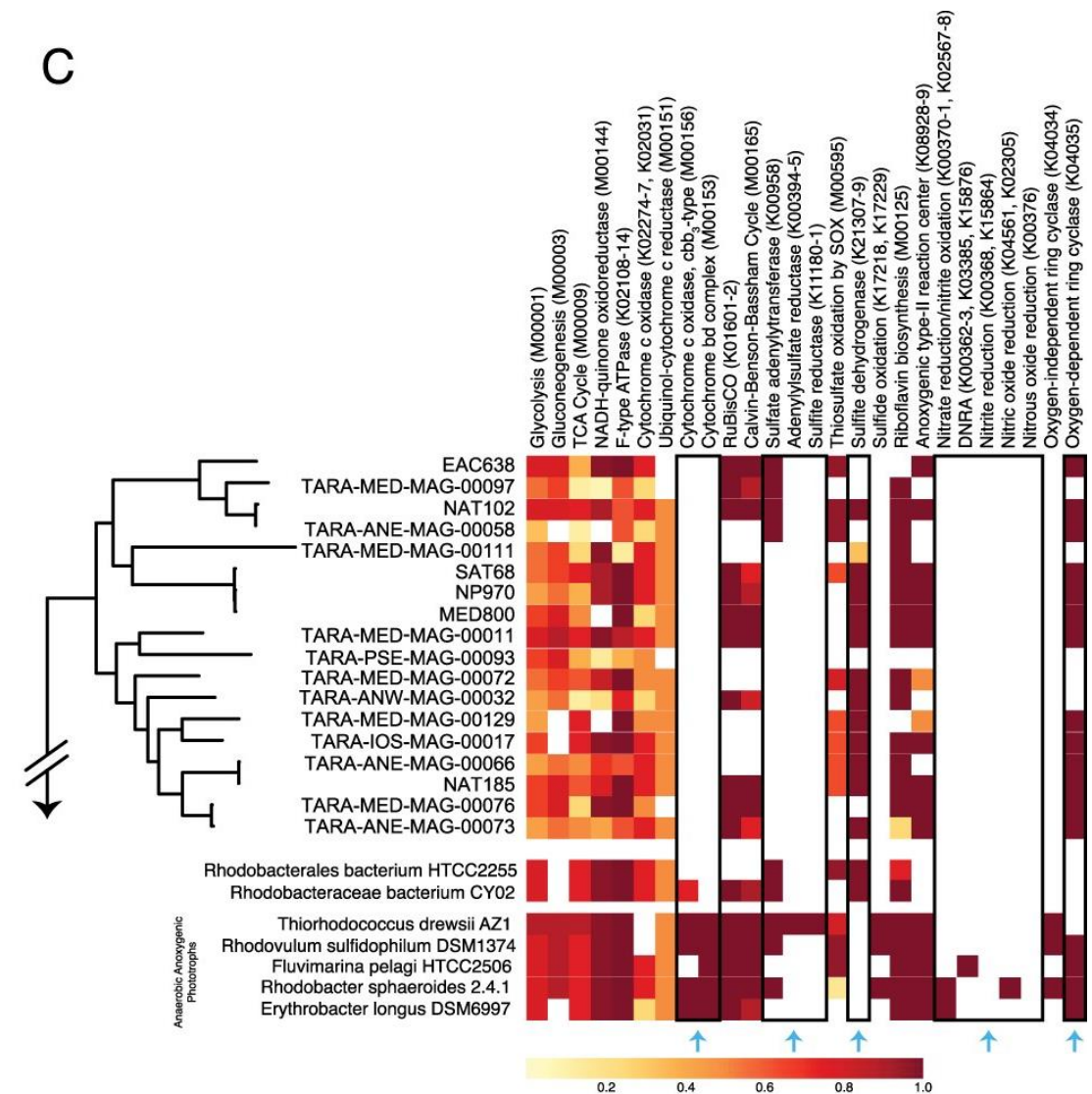
Visualization of Functional Annotation

KEGG decoder

<https://github.com/bjtully/BioData/blob/master/KEGGDecoder/README.md>

Designed to parse through a KEGG-Koala outputs (including blastKOALA, ghostKOALA, KOFAMSCAN) to determine the **completeness** of various metabolic pathways.

C



(Adapted from Graham et al. 2018)

3

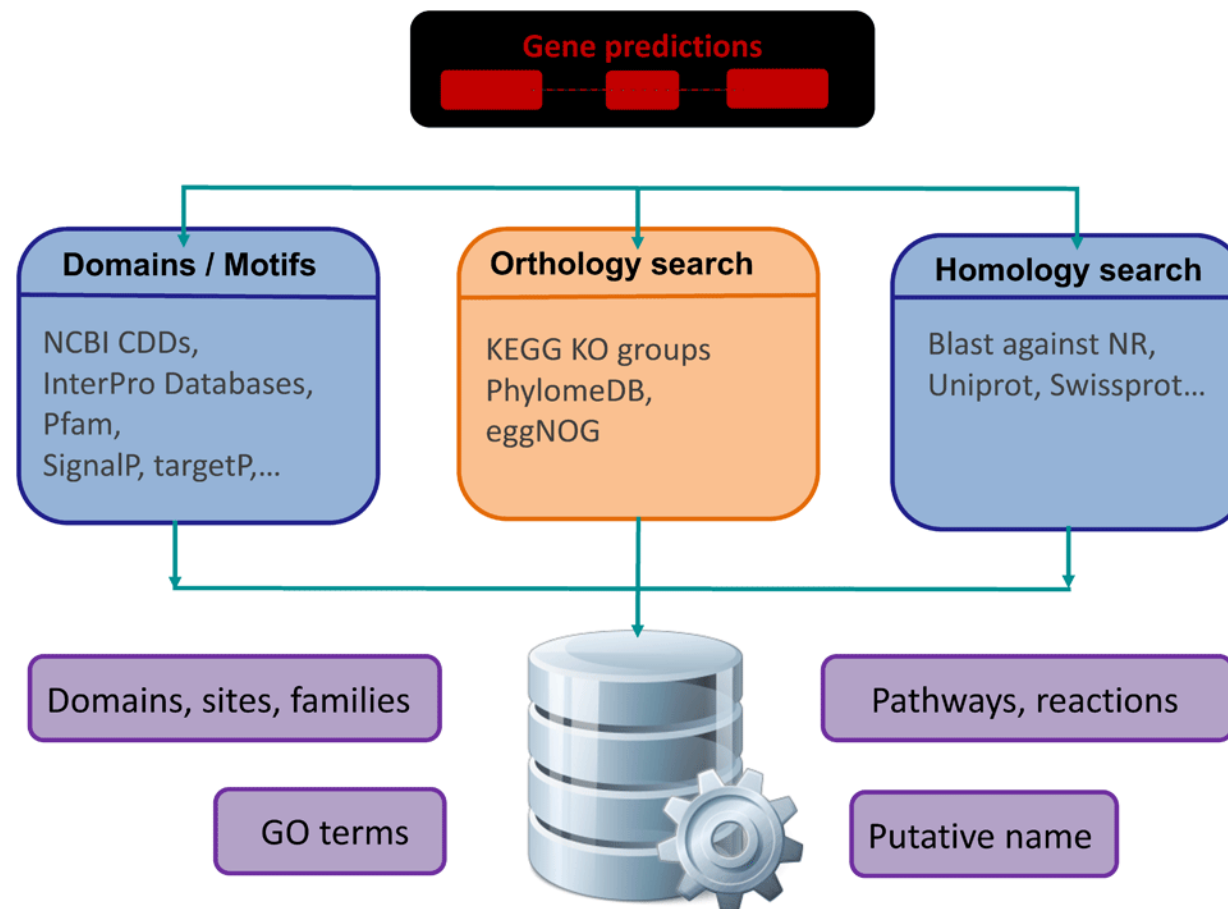
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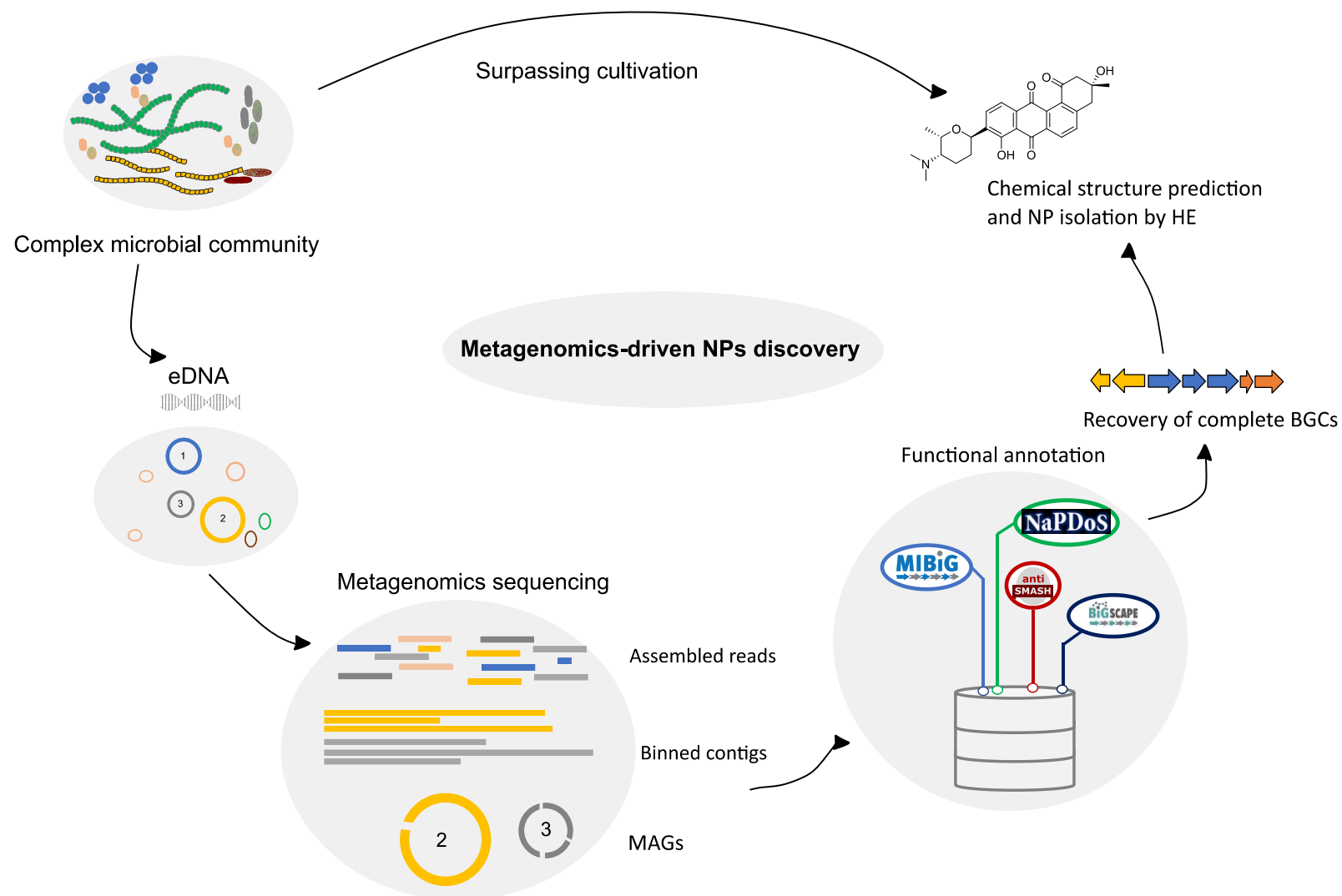
Functional metagenomics

What can they do?

Specialized metabolites



A **biosynthetic gene cluster (BGC)** can be defined as a physically clustered group of two or more genes in a particular genome that together encode a biosynthetic pathway for the production of a specialized metabolite (including its chemical variants). (Medema et al. 2015)



What can they do?

Biosynthetic gene clusters- identification



<https://antismash.secondarymetabolites.org/#!/start>

- allows the rapid genome-wide identification, annotation and analysis of secondary metabolite biosynthesis gene clusters in bacterial and fungal genomes.
- Web version and command line

```
antismash MAG_example.fa --cb-general --cb-subclusters --cb-knownclusters --  
genefinding-tool prodigal --output-dir MAG_example_antismash_output
```

4

Functional metagenomics

What can they do?

anti

SMASH

antiSMASH version 6.0.0alpha1-60bffdb

Download

About

Select genomic region:

Overview

2.1

4.1

17.1

18.1

29.1

29.2

30.1

49.1

55.1

74.1

Identified secondary metabolite regions using strictness 'relaxed'

Region	Type	From	To	Most similar known cluster	Similarity
Region 2.1	terpene	55,385	77,346		
Region 4.1	hglE-KS	1	20,725		
Region 17.1	cyanobactin	17,836	45,805	prenylagaramide B / prenylagaramide C	RiPP:Lanthipeptide 30%
Region 18.1	terpene	19,020	39,919		
Region 29.1	T1PKS, NRPS	31,802	88,579		
Region 29.2	ladderane	110,778	134,669		
Region 30.1	T3PKS	18,152	50,775		
Region 49.1	terpene	33,248	54,066		
Region 55.1	NRPS	58,824	105,581	trichamide	RiPP:Cyanobactin 18%
Region 74.1	hglE-KS, RRE-containing	43,796	71,969		

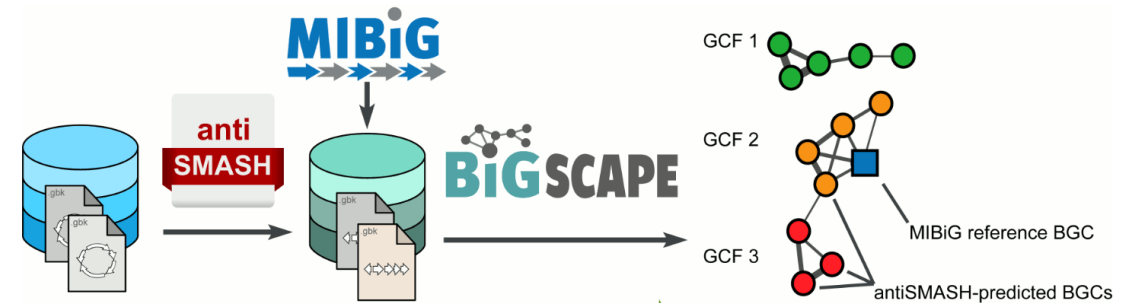
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```

4

Functional metagenomics

What can they do?

Biosynthetic gene clusters- identification



BiG-SCAPE (Biosynthetic Gene Similarity Clustering and Prospecting Engine)

- <https://bigscape-corason.secondarymetabolites.org/>

Is a software package that constructs **sequence similarity networks** of Biosynthetic Gene Clusters (BGCs) and groups them into **Gene Cluster Families (GCFs)**.

```
python bigscape.py -i antismash_gbk_files --mix --mibig --
include_singletons --cutoffs 0.3, 0.7 -o bigscape_output
```

What can they
do?

Article

**Biosynthetic potential of the global ocean
microbiome**<https://doi.org/10.1038/s41586-022-04862-3>

Received: 21 May 2021

Accepted: 12 May 2022

Published online: 22 June 2022

Open access

 Check for updates

Lucas Paoli¹, Hans-Joachim Ruscheweyh^{1,18}, Clarissa C. Forneris^{2,18}, Florian Hubrich^{2,18}, Satria Kautsar³, Agneya Bhushan², Alessandro Lotti², Quentin Clayssen¹, Guillem Salazar¹, Alessio Milanese¹, Charlotte I. Carlström¹, Chrysa Papadopoulou¹, Daniel Gehrig¹, Mikhail Karasikov^{4,5,6}, Harun Mustafa^{4,5,6}, Martin Larralde⁷, Laura M. Carroll⁷, Pablo Sánchez⁸, Ahmed A. Zayed⁹, Dylan R. Cronin⁹, Silvia G. Acinas⁸, Peer Bork^{7,10,11}, Chris Bowler^{12,13}, Tom O. Delmont^{13,14}, Josep M. Gasol⁸, Alvar D. Gossert¹⁵, André Kahles^{4,5,6}, Matthew B. Sullivan^{8,16}, Patrick Wincker^{13,14}, Georg Zeller⁷, Serina L. Robinson^{2,17}✉, Jörn Piel²✉ & Shinichi Sunagawa¹✉

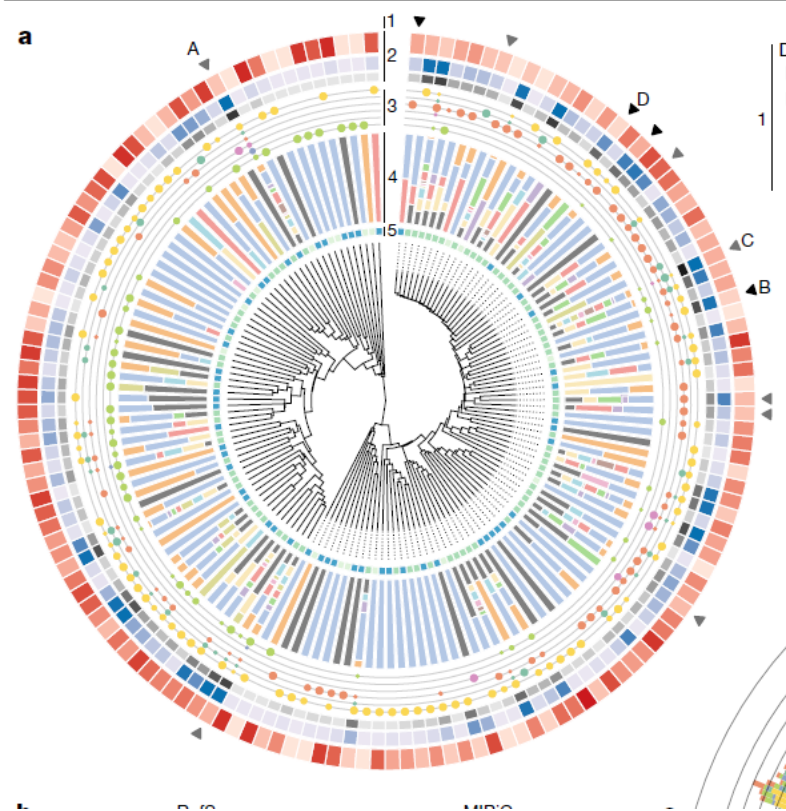
MICROBIAL GENOMICS

RESEARCH ARTICLE

Rego et al., *Microbial Genomics* 2021;7:000731

DOI 10.1099/mgen.0.000731

**Secondary metabolite biosynthetic diversity in Arctic
Ocean metagenomes**Adriana Rego^{1,2}, Antonio Fernandez-Guerra³, Pedro Duarte⁴, Philipp Assmy⁴, Pedro N. Leão^{1,*} and Catarina Magalhães^{1,5,*}

What can they
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Article

Biosynthetic potential of the global ocean microbiome

<https://doi.org/10.1038/s41586-022-04862-3>

Received: 21 May 2021

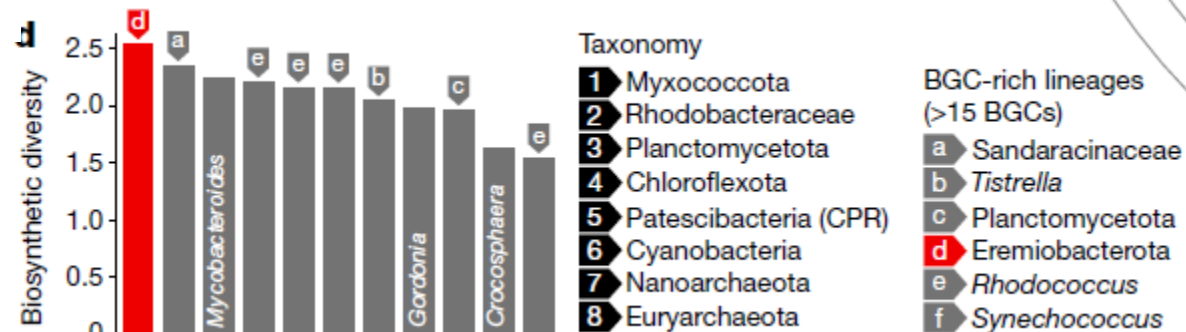
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What can they
do?

Article

Biosynthetic potential of the global ocean
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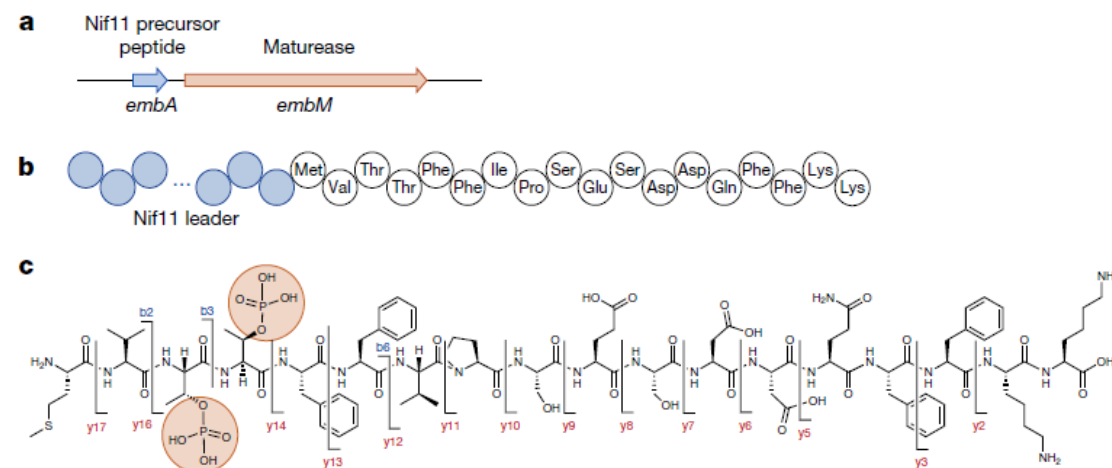
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New enzymes and natural products

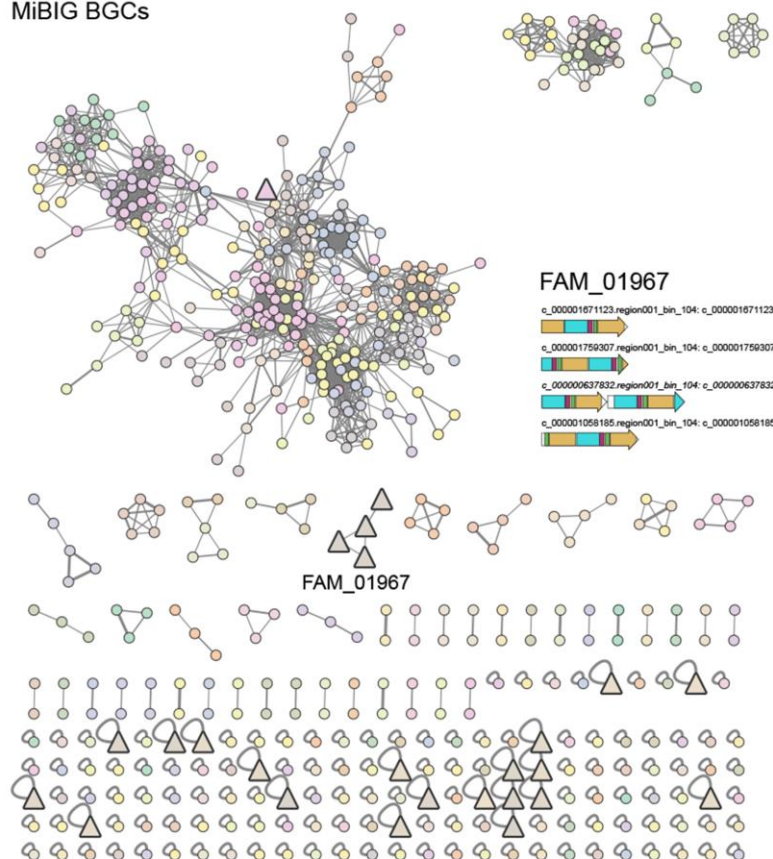
We finally sought to experimentally validate the promising prospects of our microbiomics-driven work for the discovery of new pathways, enzymes and natural products. Among the different BGC classes, RiPP pathways are known to encode a wealth of chemical and functional



What can they
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Ocean metagenomesAdriana Rego^{1,2}, Antonio Fernandez-Guerra³, Pedro Duarte⁴, Philipp Assmy⁴, Pedro N. Leão^{1,*} and Catarina Magalhães^{1,5,*}

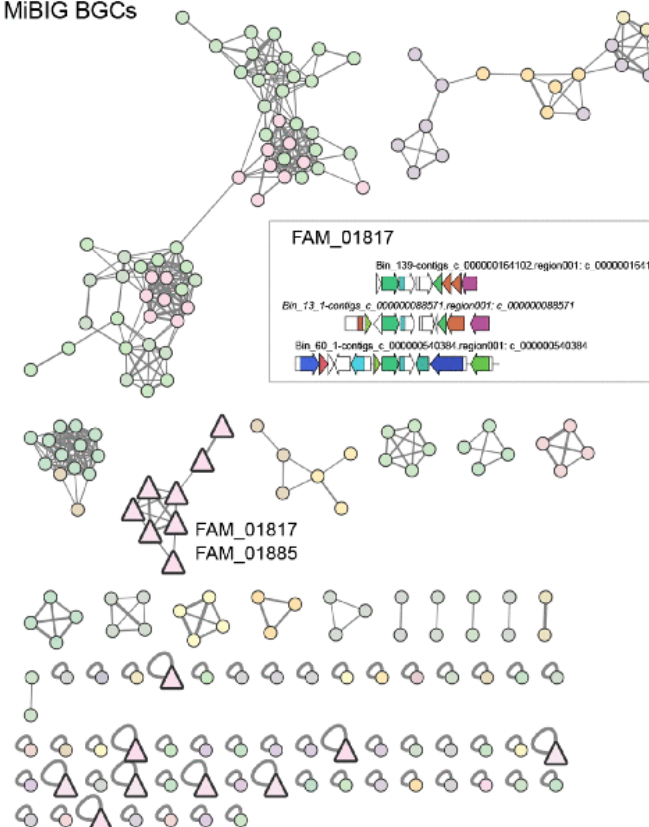
△ BGCs recovered from MAGs

○ MiBIG BGCs

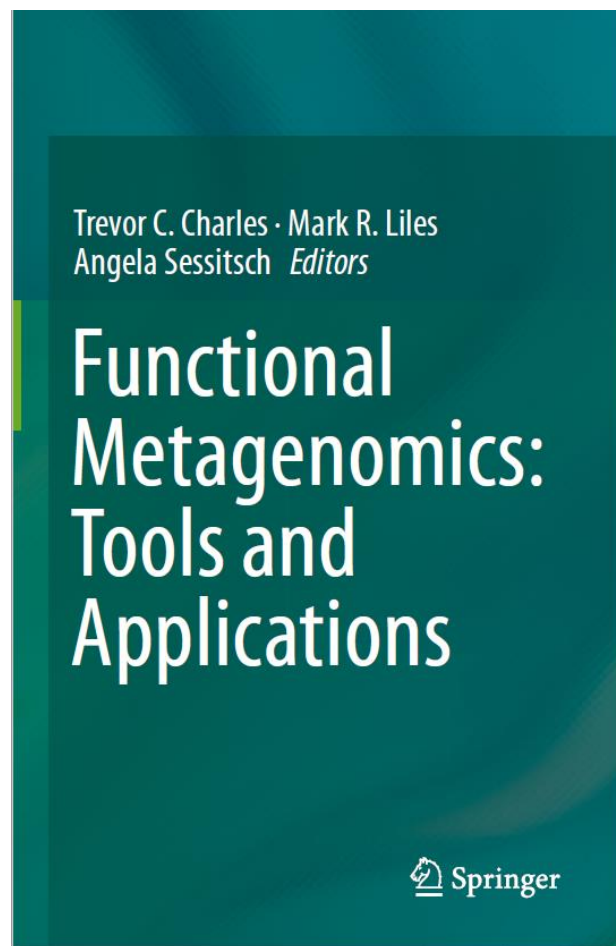


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Bibliography



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